



VECTORIZED SIGNAL STAR

A SIMD-lite extension of StarCore-1 — twice the byte throughput, one instruction.

"For when every cycle on orbit counts."

Satellites generate floods of byte-packed sensor data — temperature channels, voltage rails, GPIO status. The baseline StarCore-1 forces you to **unpack, compute, repack** every word. **VSS does both bytes at once**, with a single new opcode and a lane-split ALU. Same pipeline. Same toolchain. Half the instruction count on the workloads that hurt most.

WHAT IT DOES

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Dual 8-bit lanes

The ALU splits the carry chain at bit 7. Two independent byte ops, one clock.

+1

One new opcode

4'b1110 (VECTOR). 8 vector ops via funct[2:0]. Reserved 1010 untouched.

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Zero scalar regression

vector_mode = 0 keeps every Practical 4 instruction bit-for-bit identical.

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Drop-in datapath

Two new wires (vector_mode, funct), one new ALUOp encoding. That's the whole edit.

2x

DATA BYTES / INSTR

7x

MODELLED SPEEDUP*

SPEC SHEET

ARCHITECTURE

16-bit single-cycle RISC, Harvard

VECTOR ISA

VADD VSUB VINV VSHL VSHR VAND VOR VSLT

ENCODING

opcode 4'b1110, funct = instr[2:0]

MODES

scalar 16b / SIMD-lite 2 x 8b

MODULES TOUCHED

ALU.v · ALU_Control.v · ControlUnit.v

VERIFIED ON

Icarus Verilog 12.0 + GTKWave